Sri Lanka Institute of Information Technology

IT3021 - Data warehousing and Business Intelligence

Assignment 02

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Step 1: Data Source for the Assignment 2

1.1 Data Warehouse Design

This Hospital Emergency Room Data Warehouse (DW) is designed using a star schema to support patient flow analysis, operational monitoring, and decision-making processes. The central fact table, **Fact_ER_Visit**, captures key metrics such as wait time, satisfaction score, and visit count, and is linked to multiple dimension tables that provide contextual information.

Among these, **Dim_Department** is implemented as a Slowly Changing Dimension Type 2 (SCD Type 2) to track historical changes in department information over time. The other dimension tables include **Dim_Patient** and **Dim_Date**, each providing detailed insights into patient demographics and time-based analysis.

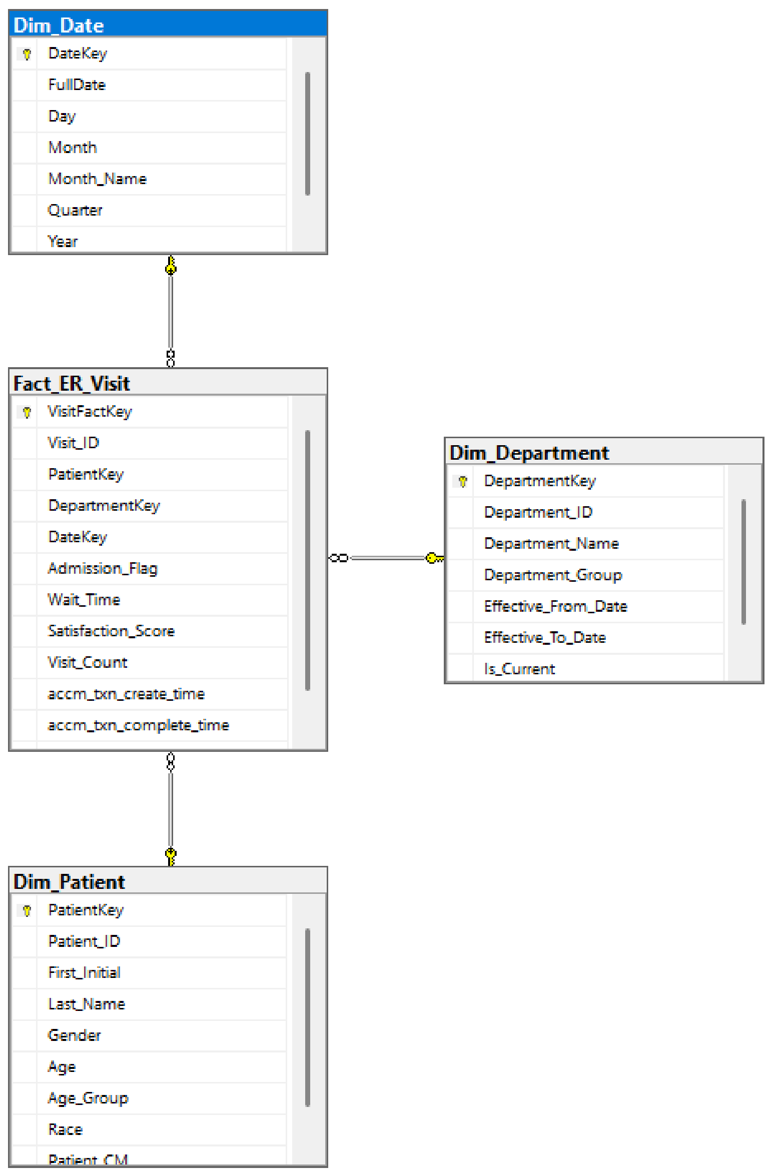


Figure 1: Star Schema of Hospital Data Warehouse

1.2 ER Diagram Representation

The Entity-Relationship (ER) diagram represents the structure of the hospital emergency room dataset used in this project.

The diagram consists of three main entities: **Patient**, **Department**, and **ER_Visit**.

- The **Patient** entity contains attributes such as Patient_ID, Gender, Age, Race, and Name details.
- The **Department** entity includes Department_ID, Department_Name, and Department_Group, representing different hospital departments.
- The **ER_Visit** entity records patient visits and includes attributes such as Visit_ID, Admission_Date, Wait_Time, Satisfaction_Score, and Admission_Flag.

The relationships between these entities are as follows:

- A **Patient** can have multiple ER visits (one-to-many relationship).
- A **Department** handles multiple ER visits (one-to-many relationship).

The ER_Visit entity acts as a central entity that connects patients and departments, capturing transactional data for each visit.

This ER diagram provides the foundation for designing the data warehouse and developing the star schema used in subsequent analysis.

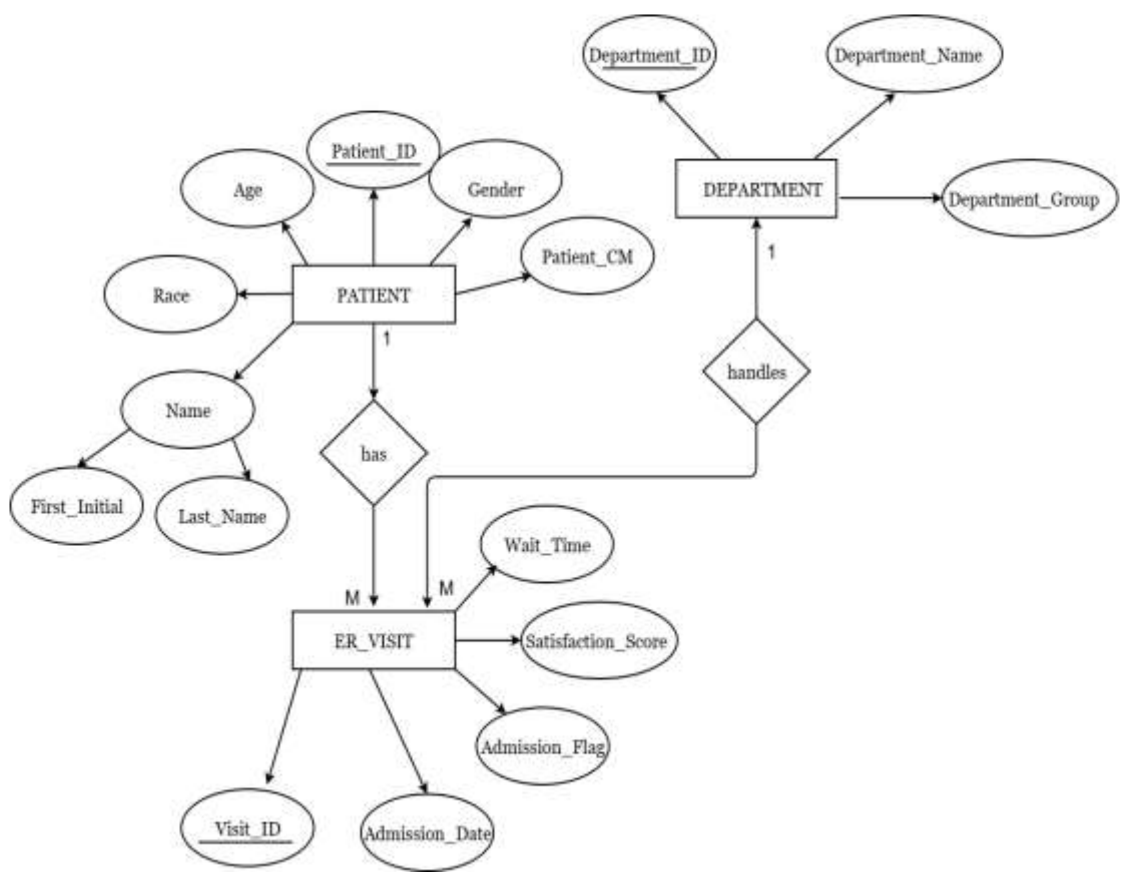


Figure 1.1 : Entity-Relationship (ER) Diagram of the Hospital Emergency Room Dataset

Step 2: SSAS Cube implementation

1. Created SSAS Project – Launched SQL Server Data Tools (SSDT) and created a new “Analysis Services Multidimensional and Data Mining Project”.
2. Established Data Source – Configured a connection to the existing hospital data warehouse database using Windows authentication.
3. Built Data Source View – Imported all fact and dimension tables from the data warehouse and established relationships between them.

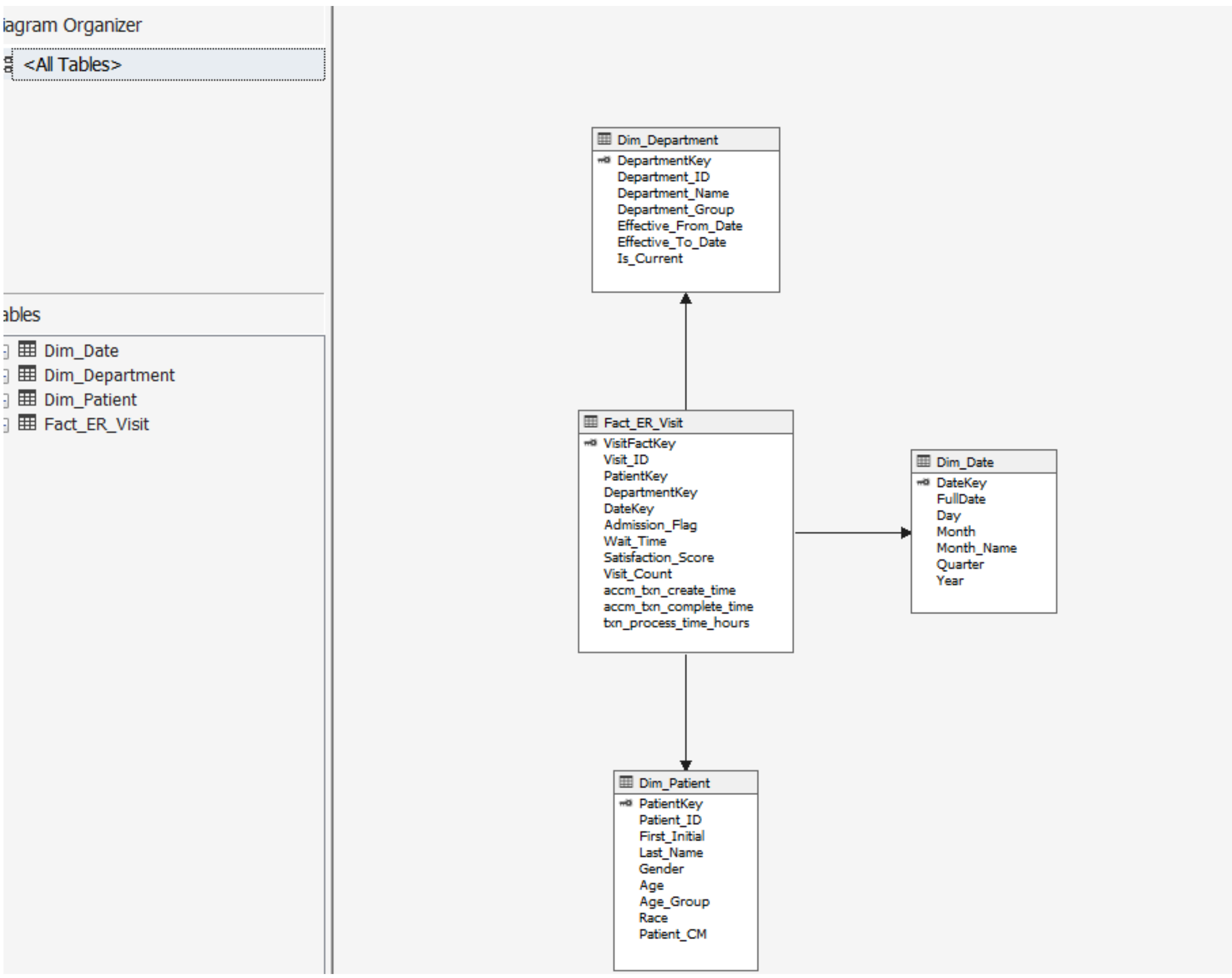


Figure 2: Data Source View (DSV) with Fact and Dimension Tables

4. Initiated Cube Creation – Used the Cube Wizard with "Use existing tables" option and selected the fact table as the measure group.
5. Incorporated Dimensions – Added all dimension tables to create the star schema structure.
6. Enhanced Dimensions – Edited each dimension to include all attributes and defined appropriate key attributes.

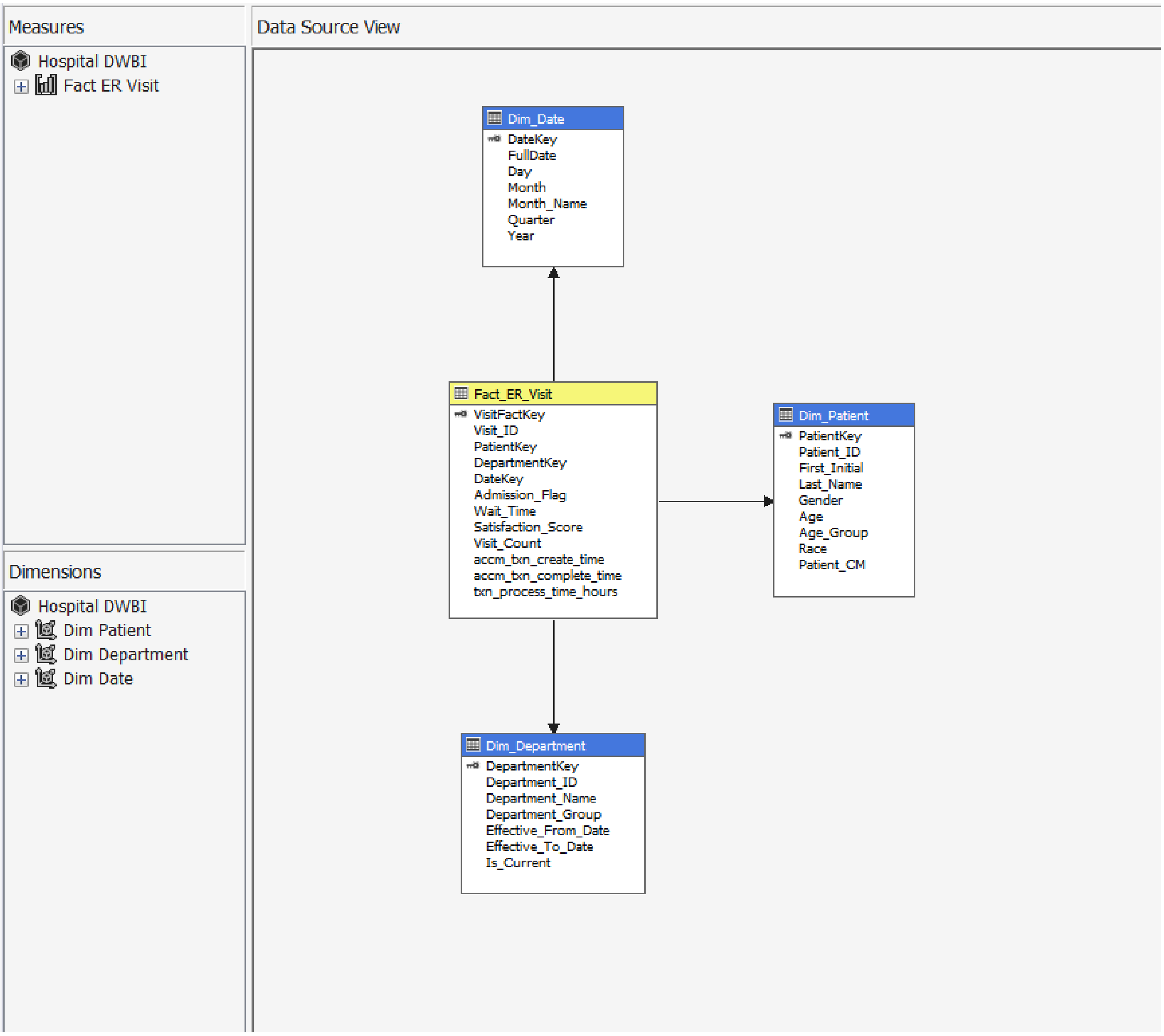


Figure 3: SSAS Cube Structure with Measures and Dimensions

7. Created Hierarchies – Built user hierarchies:

- Calendar Hierarchy → Year – Quarter – Month – Day

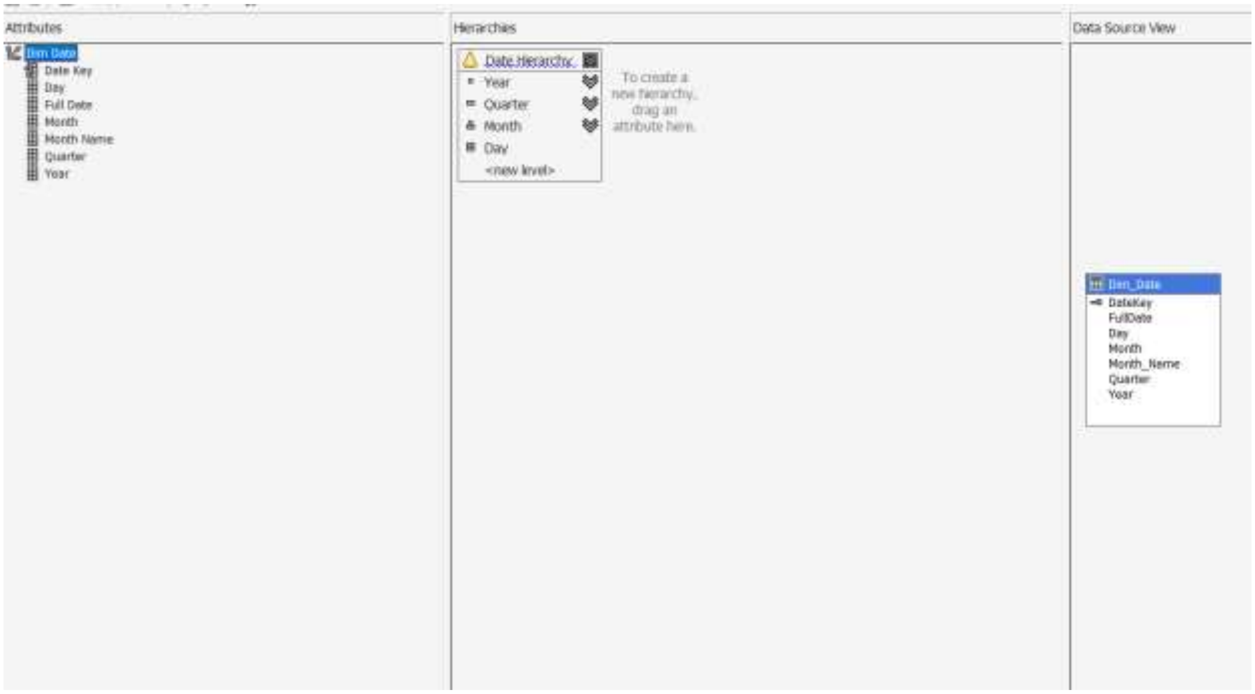


Figure 4: Hierarchy Creation (Date Hierarchy)

Department Hierarchy → Department Group – Department Name

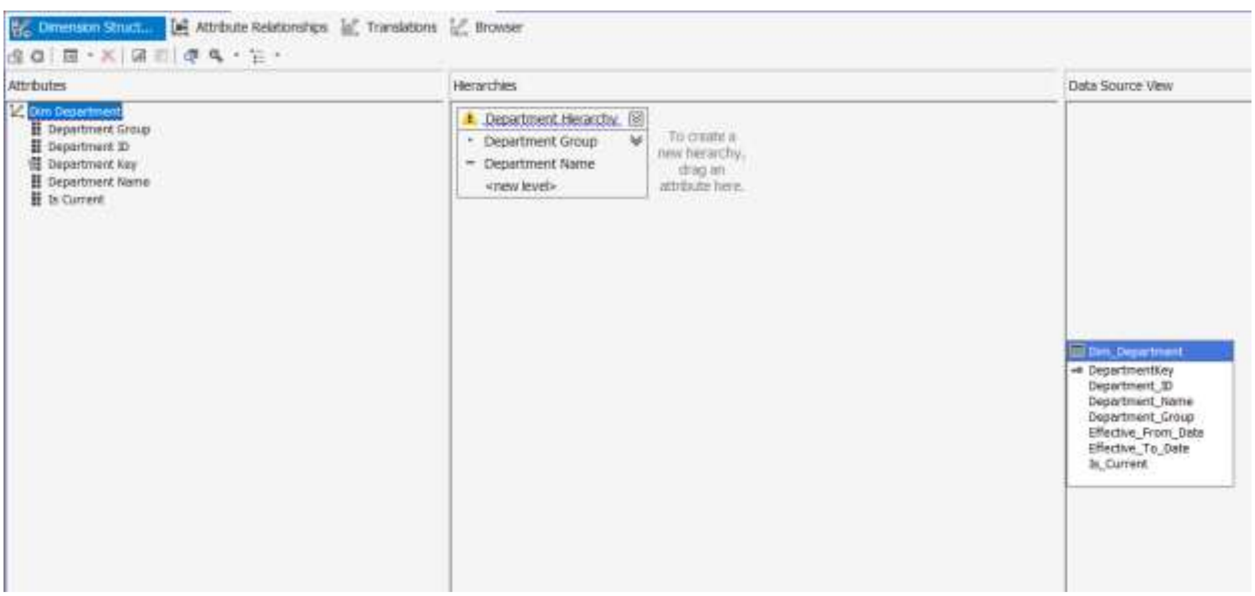


Figure 5: Dimension Structure (Dim_Department)

8. Verified Relationships – Checked and corrected dimension usage mappings between fact and dimension tables.

9. Deployed the Cube – Configured deployment settings and successfully processed the cube to the SSAS server.
10. Validated Deployment – Confirmed cube accessibility through SSDT browser and SQL Server Management Studio.

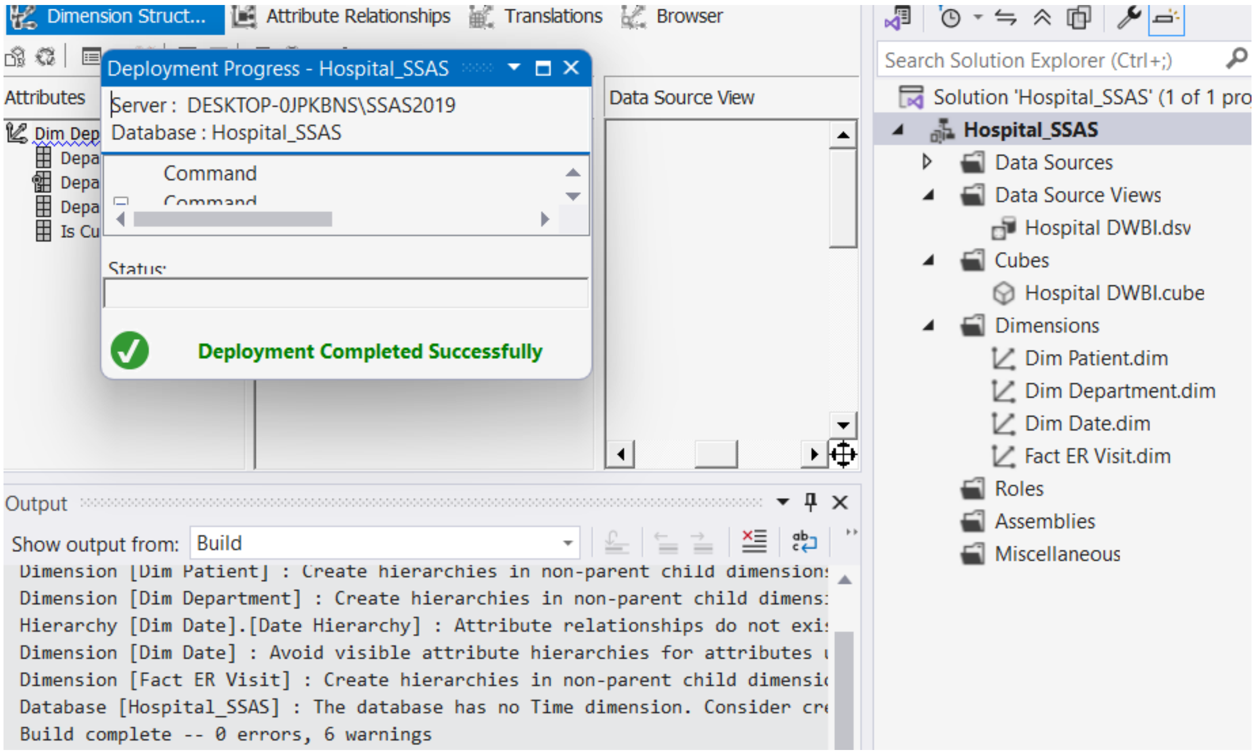


Figure 6: Cube Deployment Success

Step 3: Demonstration of OLAP operations

1. Connected Excel to SSAS Cube – Used Excel's Get Data feature to establish a connection to the deployed cube.
2. Created PivotTable – Generated a PivotTable report for OLAP analysis.
3. **Demonstrated Roll-up:**

- Roll-up operation was performed to aggregate data from lower levels to higher levels in the hierarchy. In this case, visit count data was aggregated from the quarterly level to yearly totals, providing a summarized view of the data for easier analysis and comparison.

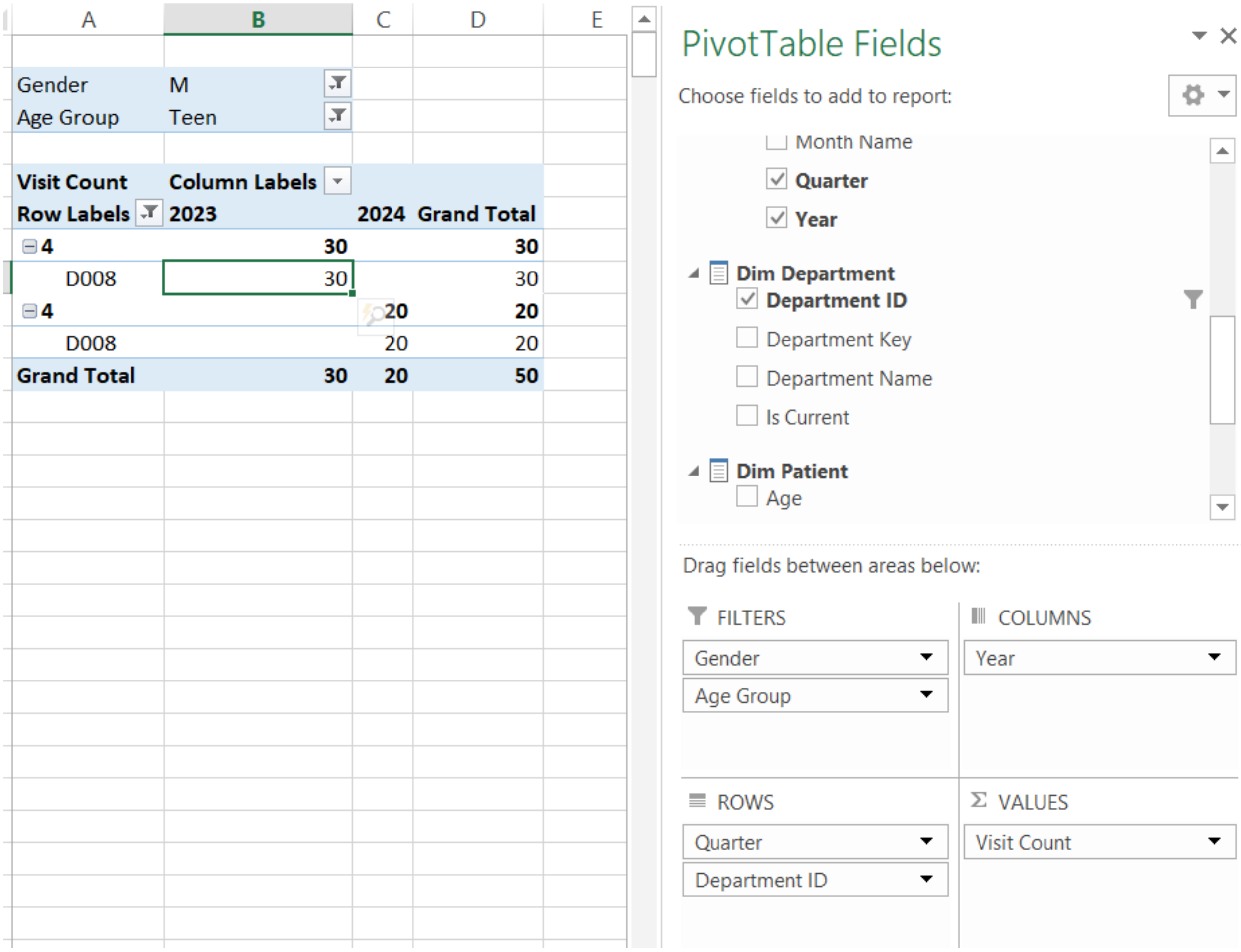


Figure 7: Excel Connection to SSAS Cube

8. Performed Drill-down

- Drill-down operation was performed by expanding hierarchy levels to analyze data at a more detailed level. The data was explored from Year to Quarter, allowing deeper insight into how values are distributed across different time periods.

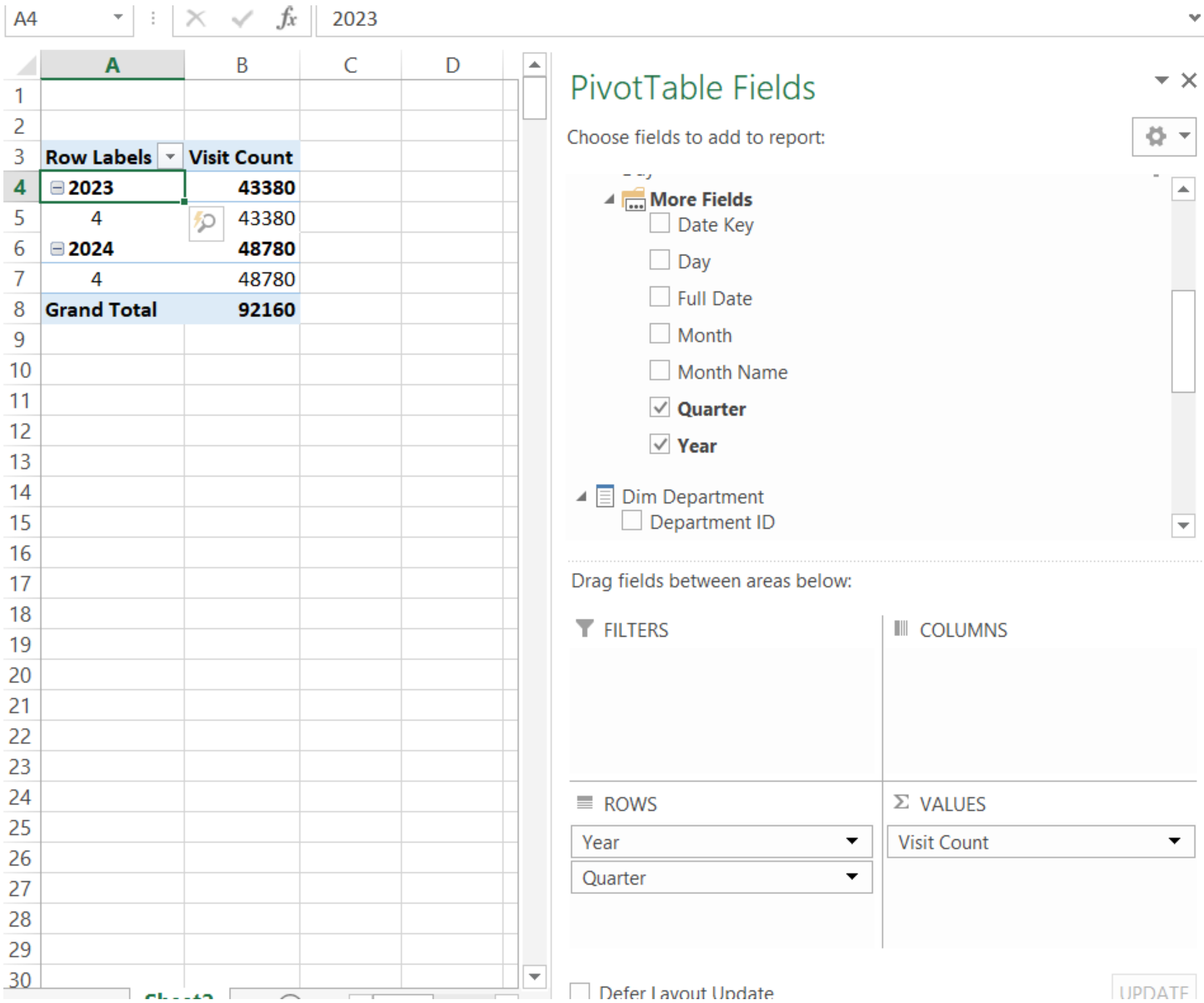


Figure 8: PivotTable Fields and Data View

9. Applied Slice Operation

- Slice operation was used to filter data based on specific conditions to obtain a focused view. In this case, the data was filtered by Gender (M) and Age Group (Teen), enabling analysis of a specific subset of the dataset.

Gender	M
Age Group	Teen

Visit Count	Column Labels		
Row Labels	2023	2024	Grand Total
4	30		30
D008	30		30
4		20	20
D008		20	20
Grand Total	30	20	50

Figure 9: Slice Operation (Filtered Data)

10. Executed Dice Operation

Dice operation was performed by applying multiple filters simultaneously to create a sub-cube. The data was filtered based on Gender (M), Age Group (Teen), and Department ID (D008), allowing detailed analysis of a specific segment of the data

Gender	M
Age Group	Teen
Department ID	D008

Row Labels	Visit Count
2023	30
4	30
2024	20
4	20
Grand Total	50

Figure 10: Dice Operation (Multiple Filters Applied)

10. Showed Pivot Functionality

Pivot operation was used to reorganize and display data by arranging Year in columns and Quarter and Department in rows. This allowed comparison of values across different dimensions and improved the overall readability of the data.

	A	B	C	D
1				
2	Gender	M		
3	Age Group	Teen		
4				
5	**Visit Count**	**Column Labels**		
6	**Row Labels**	**2023**	**2024**	**Grand Total**
7	**4**	**30**		**30**
8	D008	30		30
9	**4**		**20**	**20**
10	D008		20	20
11	**Grand Total**	**30**	**20**	**50**
12				
13				

Figure 11: Pivot Operation (Rows and Columns Swapped)

Step 4 : PowerBI Reports

4.1 Connected to Data Source

Power BI Desktop was connected to the SQL Server Analysis Services (SSAS) cube to import the fact and dimension tables for reporting.

2. The star schema was already linked

The star schema was already defined in the cube, where the central fact table **Fact_ER_Visit** is connected with the dimension tables **Dim_Patient, Dim_Department, and Dim_Date**. These relationships enable efficient analysis of emergency room operations.

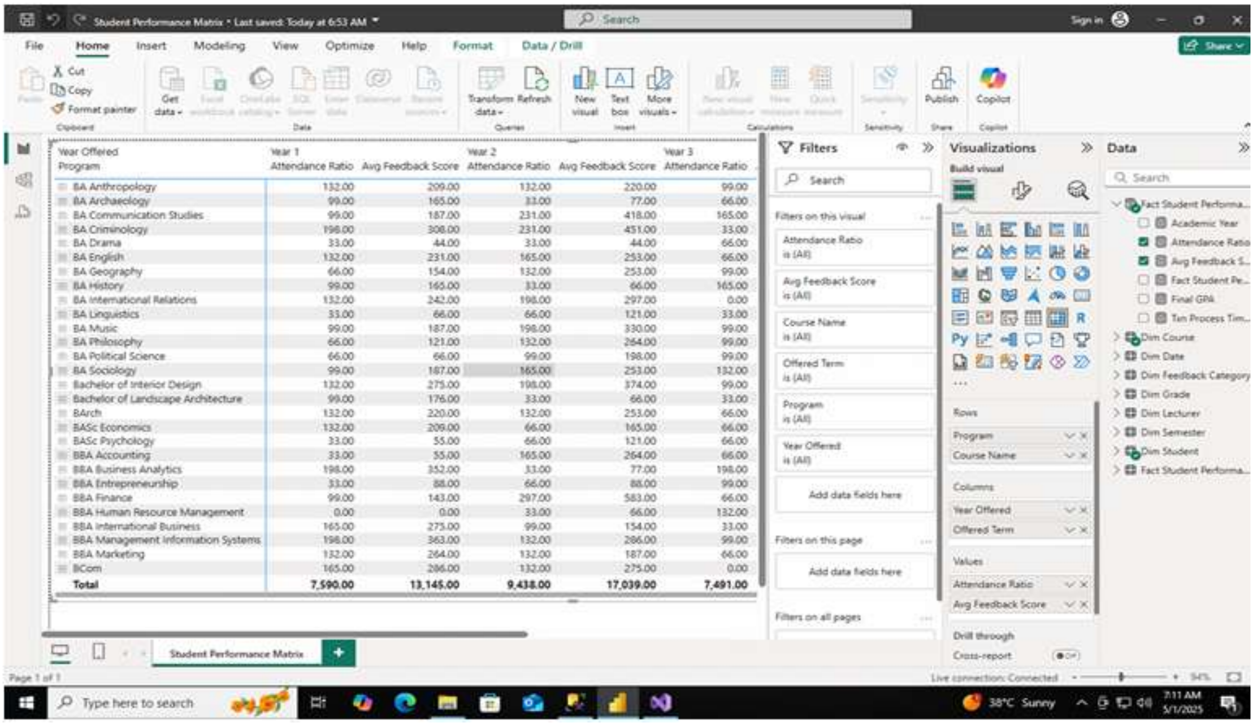
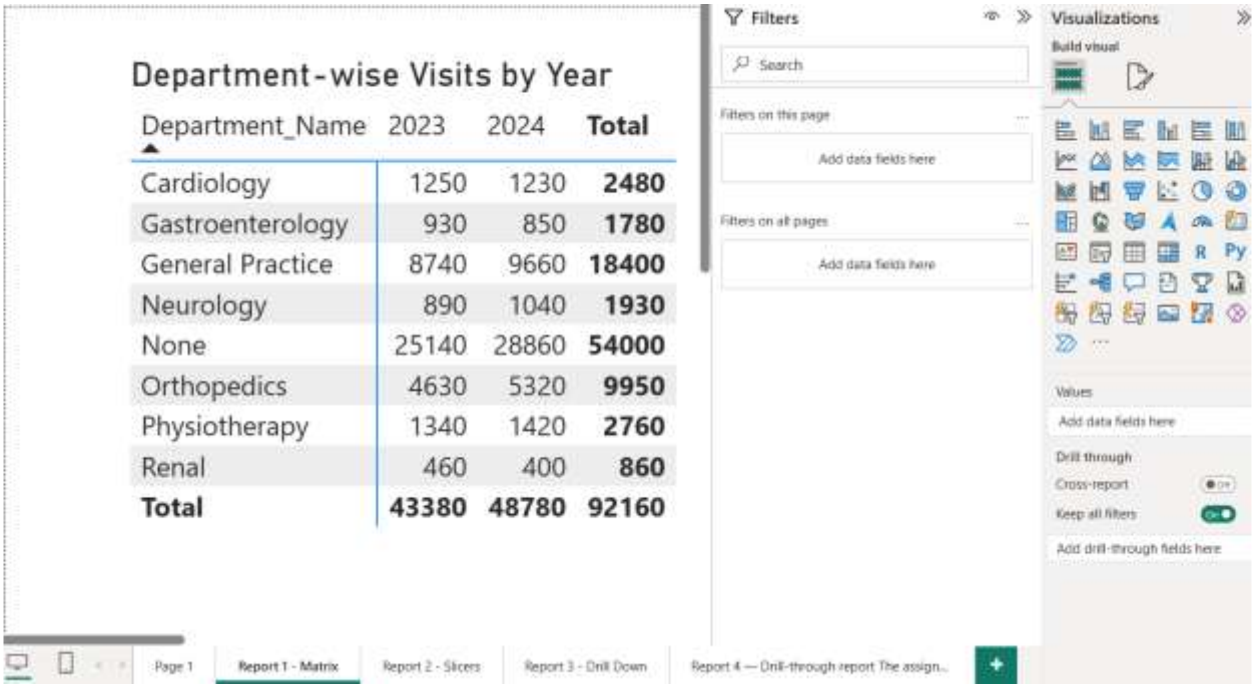


Figure 12: Power BI Model View (Star Schema Relationships)

4.2 Report 1:Matrix Visual

Designed Matrix Layout:

The matrix visual was configured by assigning fields to rows, columns, and values to display department-wise visit data across different years. The data was grouped and filtered based on specific conditions to enable focused analysis. Totals and subtotals were enabled to summarize the data, and basic formatting such as alignment and column width adjustments was applied to improve readability and presentation.



Department_Name	2023	2024	Total
Cardiology	1250	1230	**2480**
Gastroenterology	930	850	**1780**
General Practice	8740	9660	**18400**
Neurology	890	1040	**1930**
None	25140	28860	**54000**
Orthopedics	4630	5320	**9950**
Physiotherapy	1340	1420	**2760**
Renal	460	400	**860**
Total	**43380**	**48780**	**92160**

Figure 13: Matrix Visual – Department-wise Visits by Year

4.3 Report 2 : Multiple Slicers with Cascading Filters

a) Created Interactive Filters:

Two slicers were created to enable interactive filtering of the dashboard:

- **Year**
- **Quarter**

These slicers allow users to filter emergency room data dynamically based on time. Selecting a specific year and quarter updates all visuals accordingly, enabling focused analysis of patient visits.

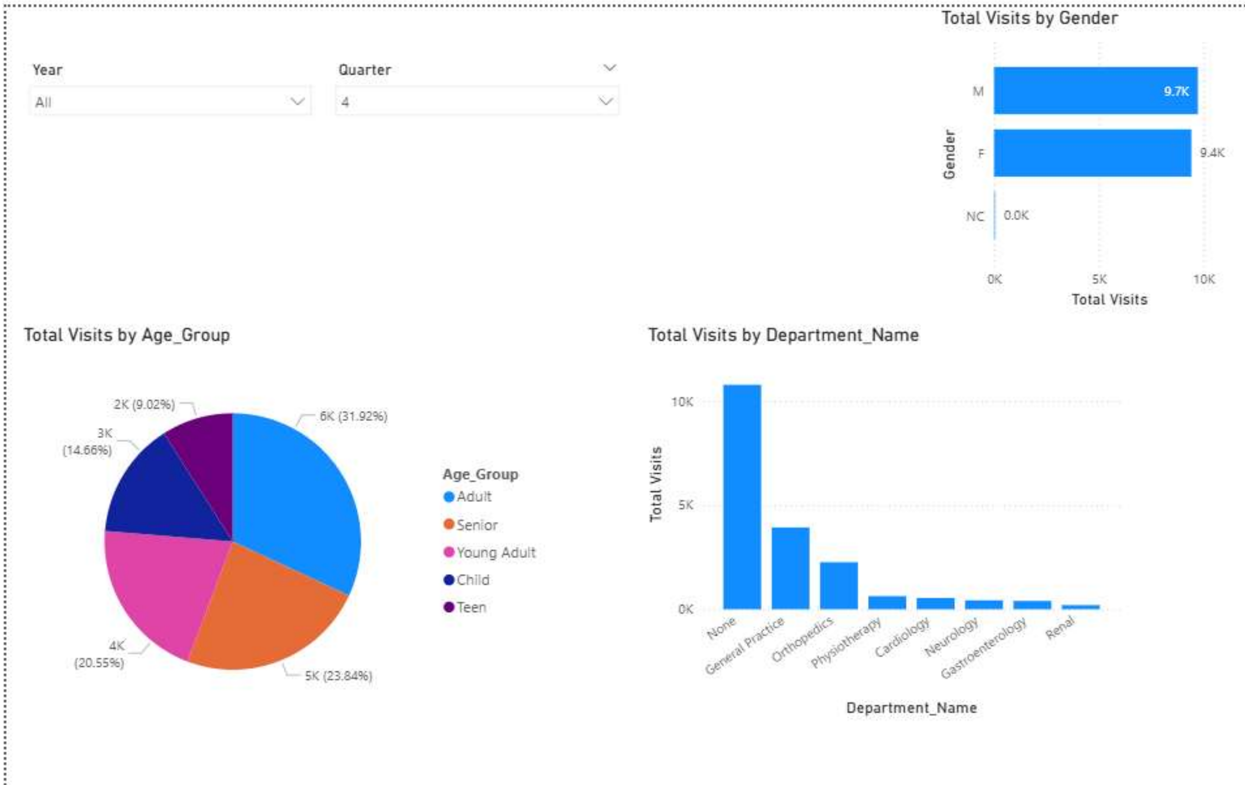


Figure 14: Initial Dashboard View with Slicers

b) Added Supporting Visuals:

Several visuals were created to analyze emergency room performance:

- **Bar Chart:** Total Visits by Gender
- **Pie Chart:** Total Visits by Age Group
- **Column Chart:** Total Visits by Department Name

These visuals provide insights into patient distribution across gender, age groups, and hospital departments.

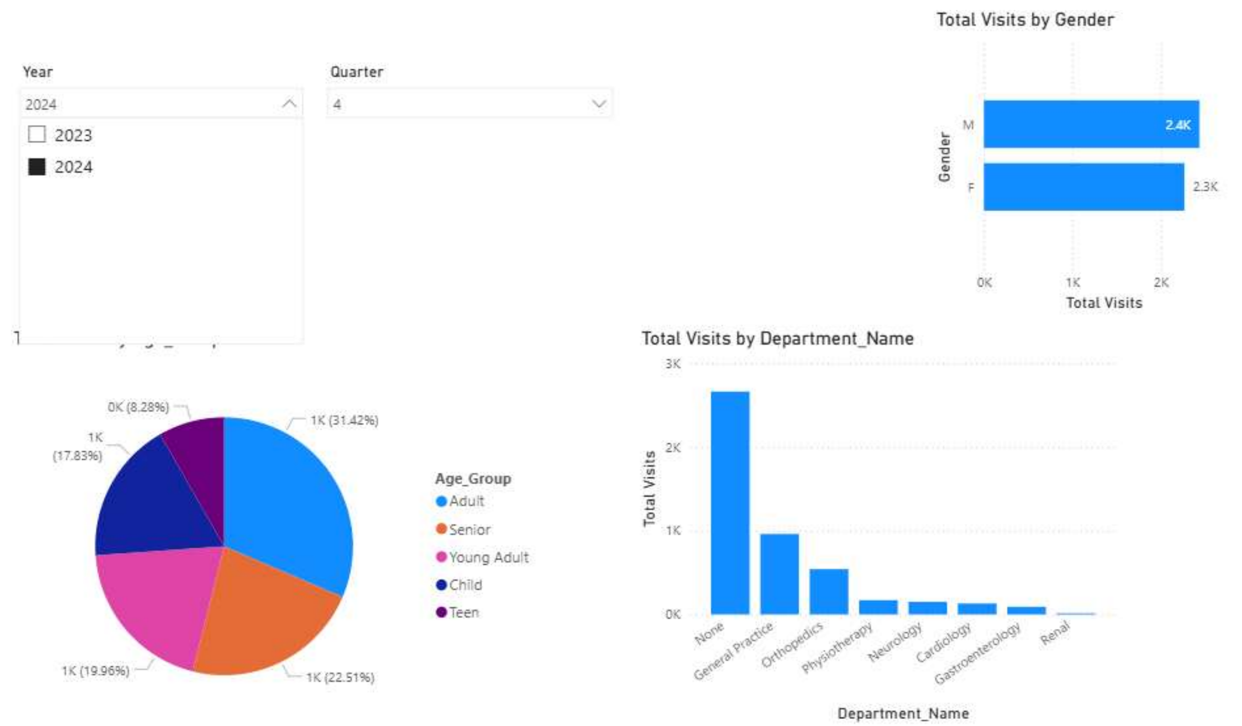


Figure 15: Dashboard with Year Filter Applied

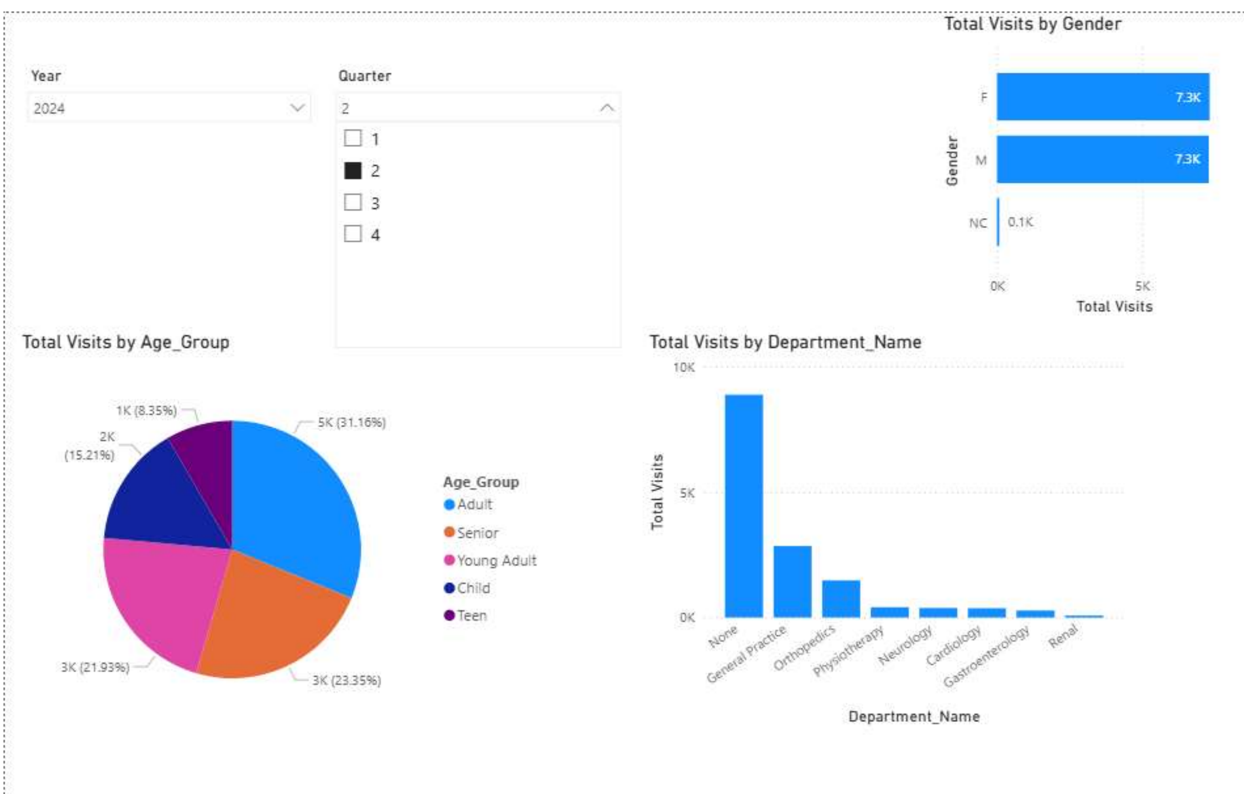


Figure 16: Dashboard with Quarter Filter Applied

c) Cascading Filter Behavior:

The slicers work together in a cascading manner. When a year is selected, the available quarters are filtered accordingly. Similarly, selecting a quarter refines the dataset further.

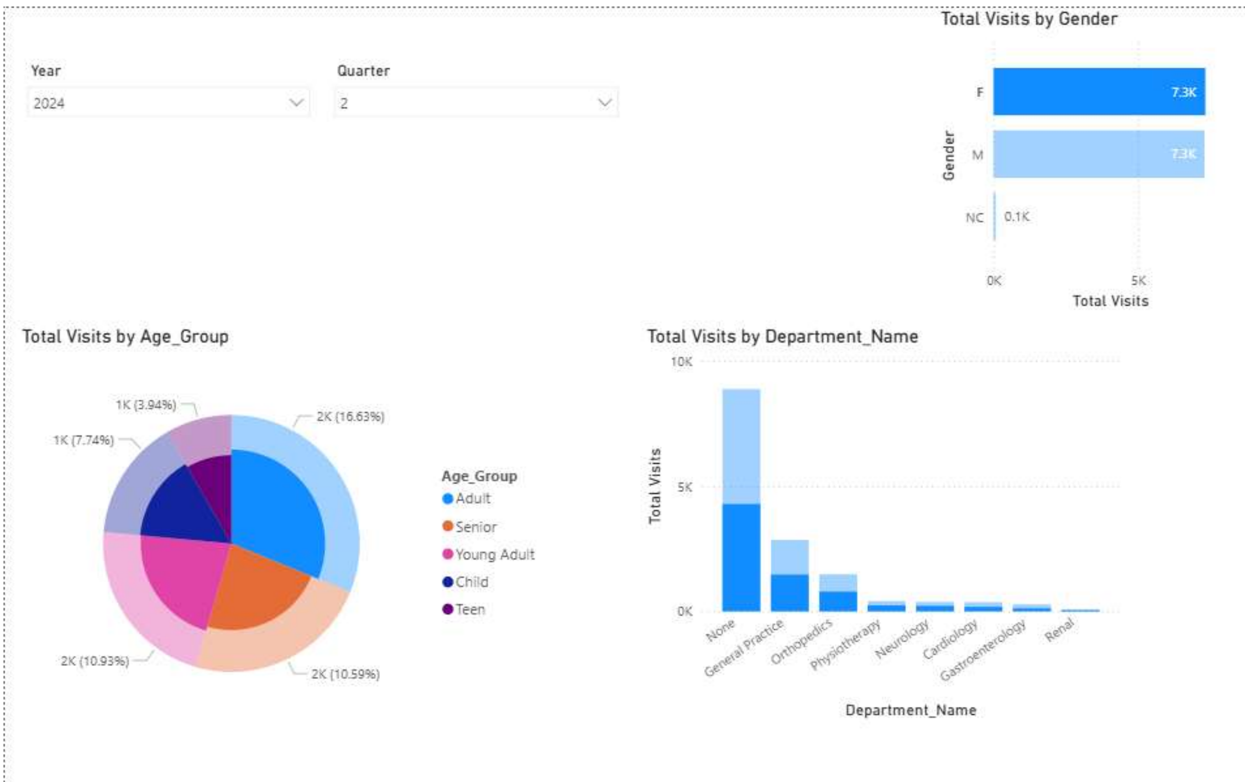


Figure 17: Dashboard with Combined Filters

d) Test the Filters:

The filters were tested by selecting different values for Year and Quarter. It was observed that:

- All visuals updated dynamically
- Data changed consistently across all charts
- The dashboard responded correctly to user inputs

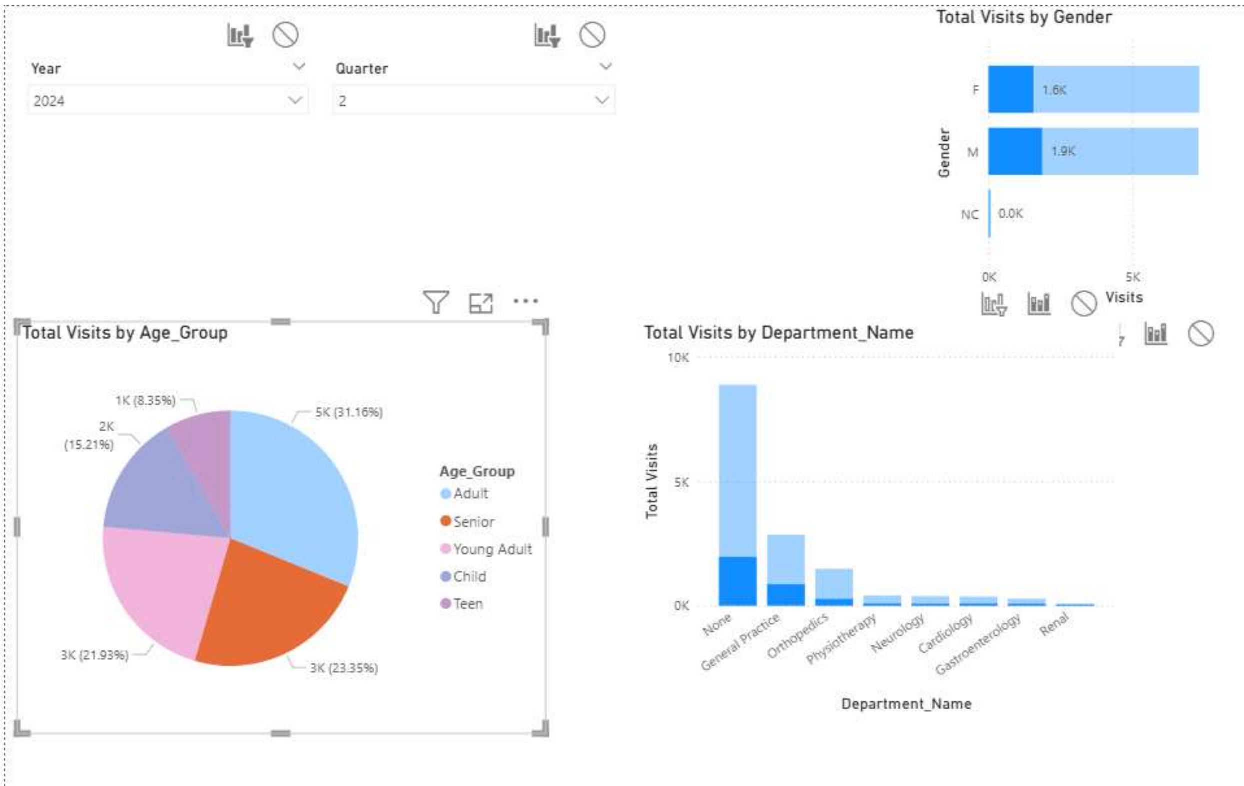


Figure 18: Filter Interaction Result

4.4 Report 3: Drill Down Report

a) Built Hierarchical Visual:

A hierarchical visual was created using the **Date dimension** to enable drill-down analysis of patient visits across different time levels.

The hierarchy includes:

- **Year → Quarter → Month**

This structure allows users to analyze data from a high-level summary down to detailed levels.

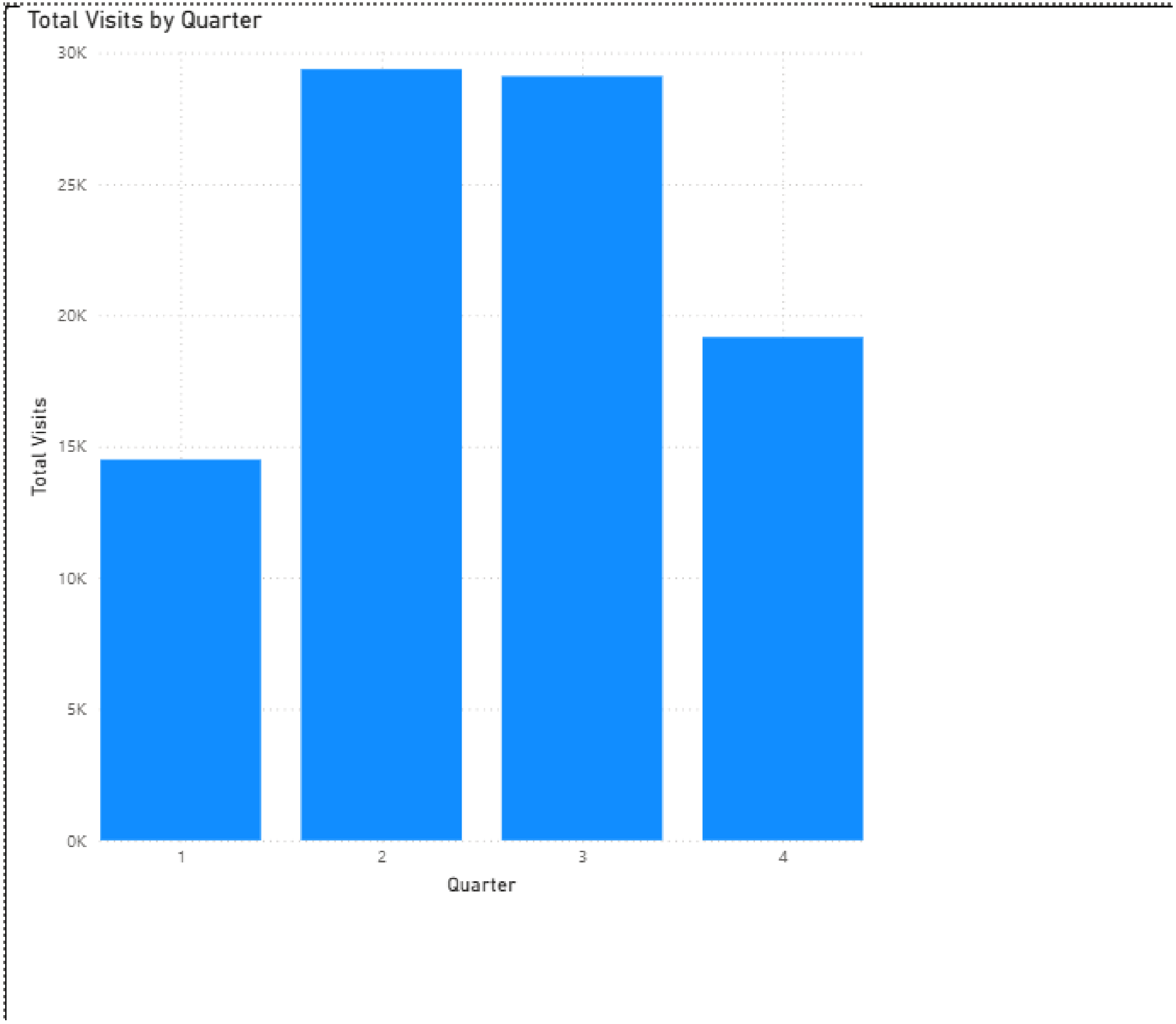


Figure 19: Year Level View (Total Visits by Year)

b) Added visuals:

A column chart was used to represent total visits across the hierarchy levels.

- X-axis → Time hierarchy (Year / Quarter / Month)
- Y-axis → Total Visits

This visual allows users to interactively explore data across different time levels.

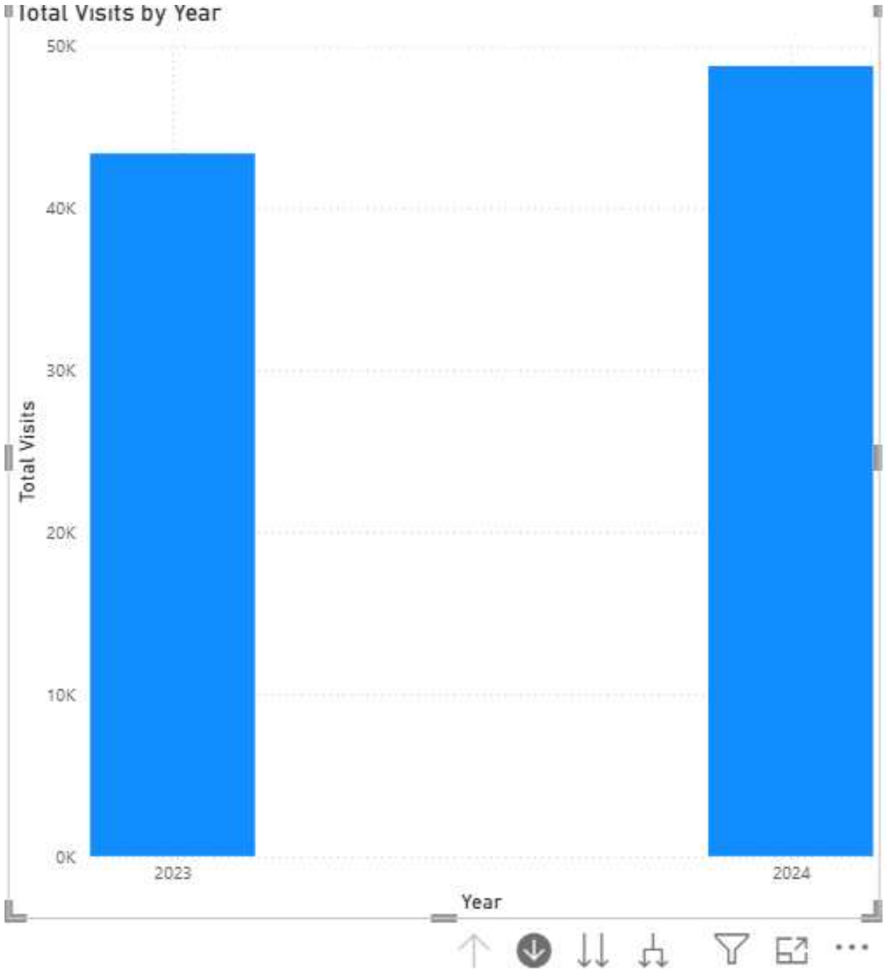


Figure 20: Quarter Level View (Drill-down)

c) Set up the drill-down functionality:

Drill-down functionality was enabled using Power BI's hierarchy feature. Users can navigate through levels by clicking the drill-down buttons.

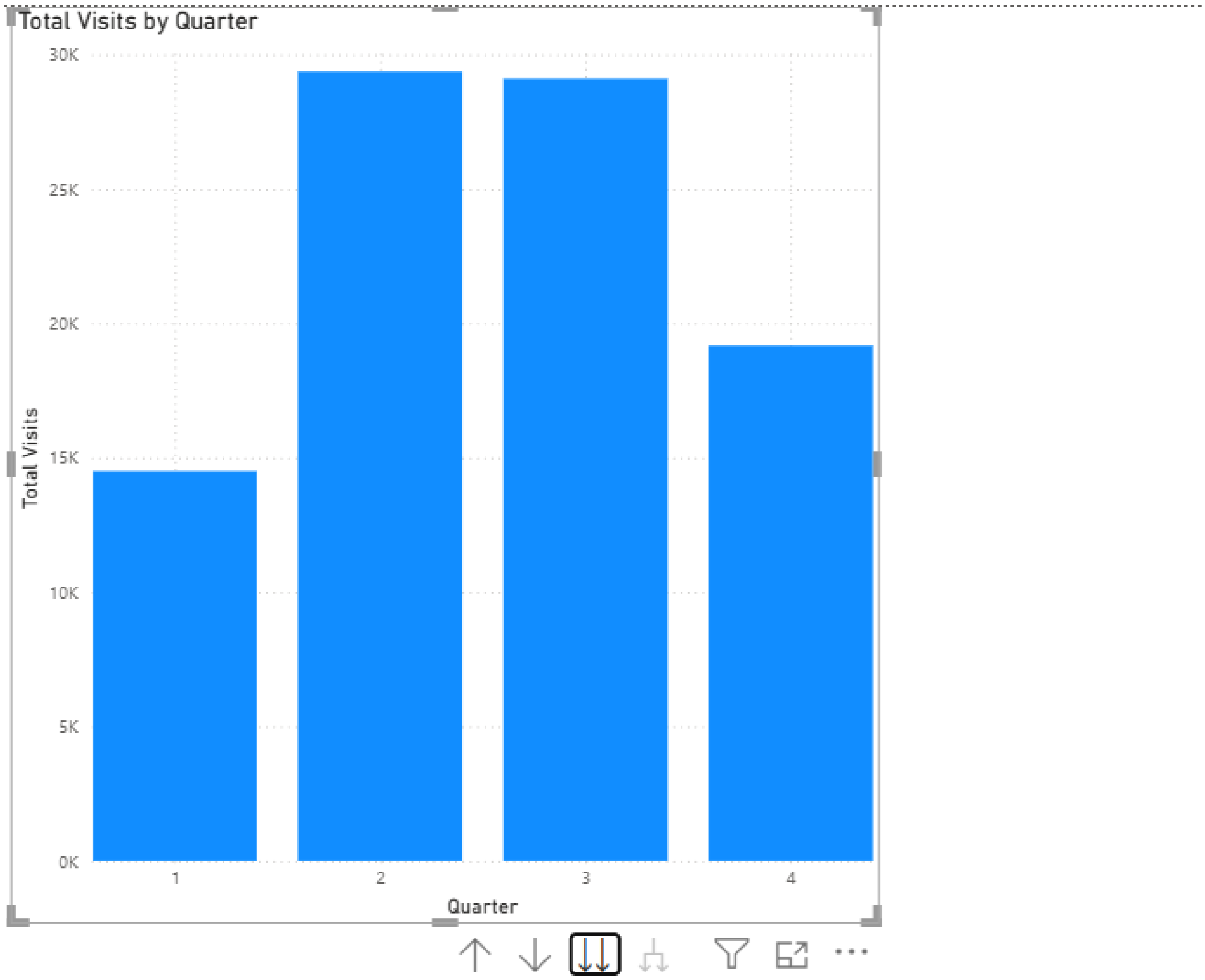


Figure 21: Drill-down Action using Hierarchy

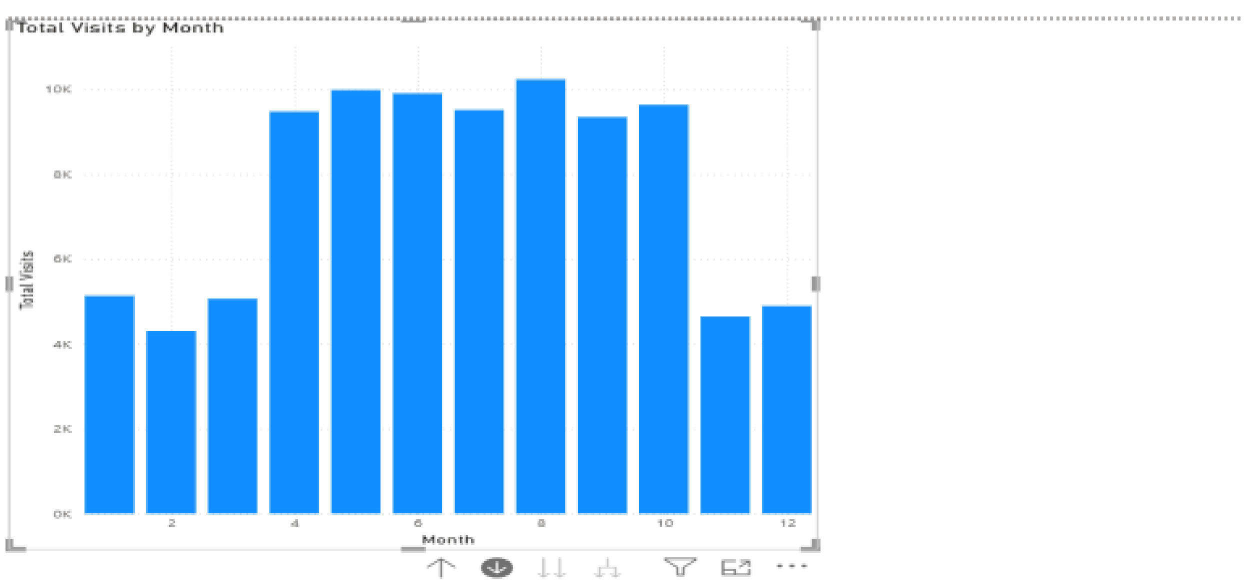


Figure 22: Month Level View (Detailed Analysis)

Report 4: Drill-Through Report

a) Created Overview Page:

An overview report page was created to display summary-level information about emergency room visits.

This page includes visuals such as:

- Total Visits by Department
- Patient distribution by Age Group
- Total Visits by Gender

This page allows users to select specific data points and navigate to detailed views.

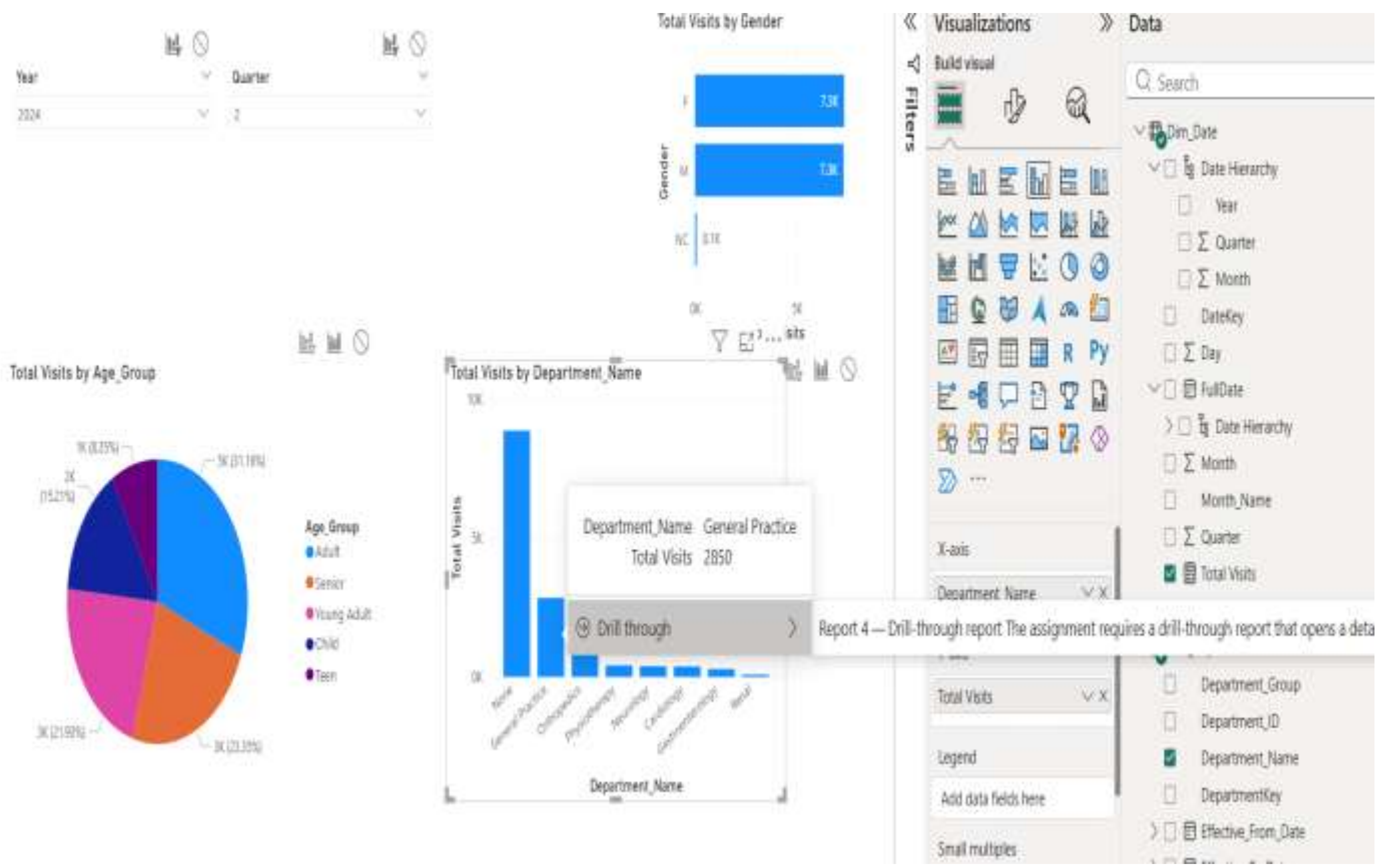


Figure 23: Drill-through Option (Right-click Menu)

b) Created Detail Page:

A detailed report page was created to display in-depth information for the selected data.

This page includes:

- Card visual → Total Visits
- Table → Patient details (Gender, Age Group, Visit Count)
- Bar chart → Total Visits by Gender

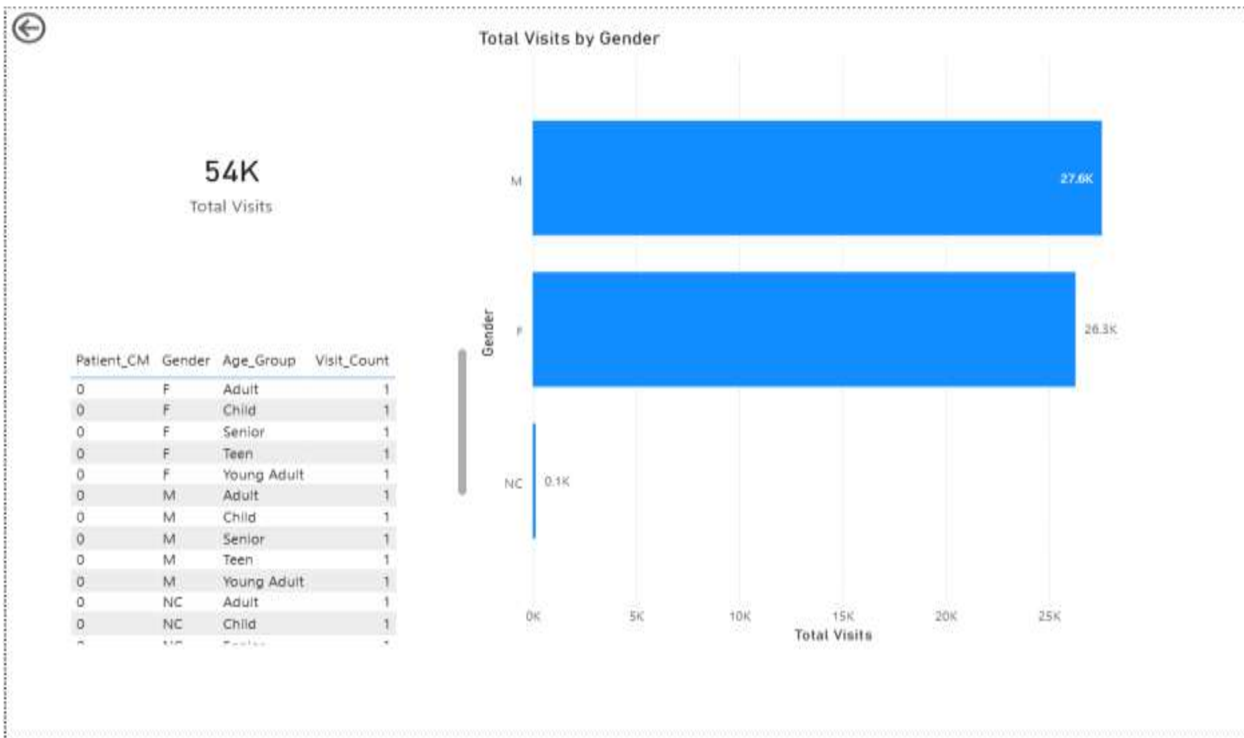


Figure 24: Detail Page (Full Data View)

c) Set up Drill-through:

Drill-through functionality was configured by adding the field:

- **Department Name**

to the **Drill-through filter section** in Power BI.

This ensures that when a user selects a department from the overview page, the detail page displays only relevant data.

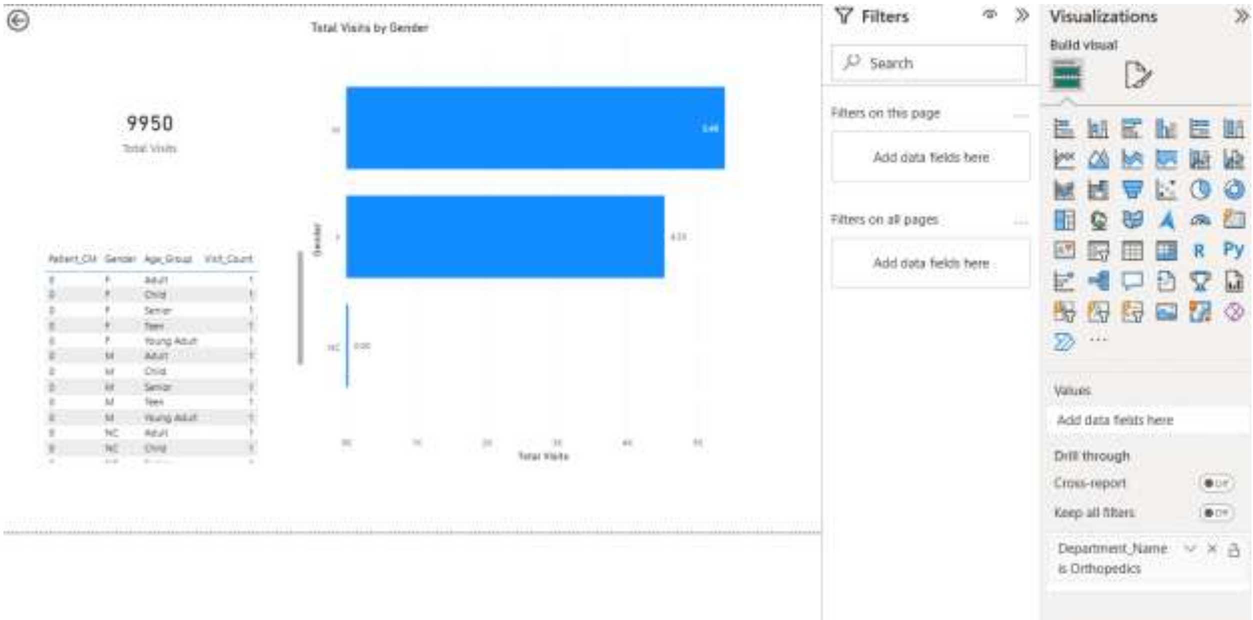


Figure 25: Drill-through Applied (Filtered by Department)

d) Test Drill-through:

The drill-through functionality was tested by:

1. Right-clicking on a department in the overview page
2. Selecting the drill-through option
3. Navigating to the detail page

The results confirmed that:

- Data is filtered correctly
- All visuals update dynamically
- The selected department's details are displayed

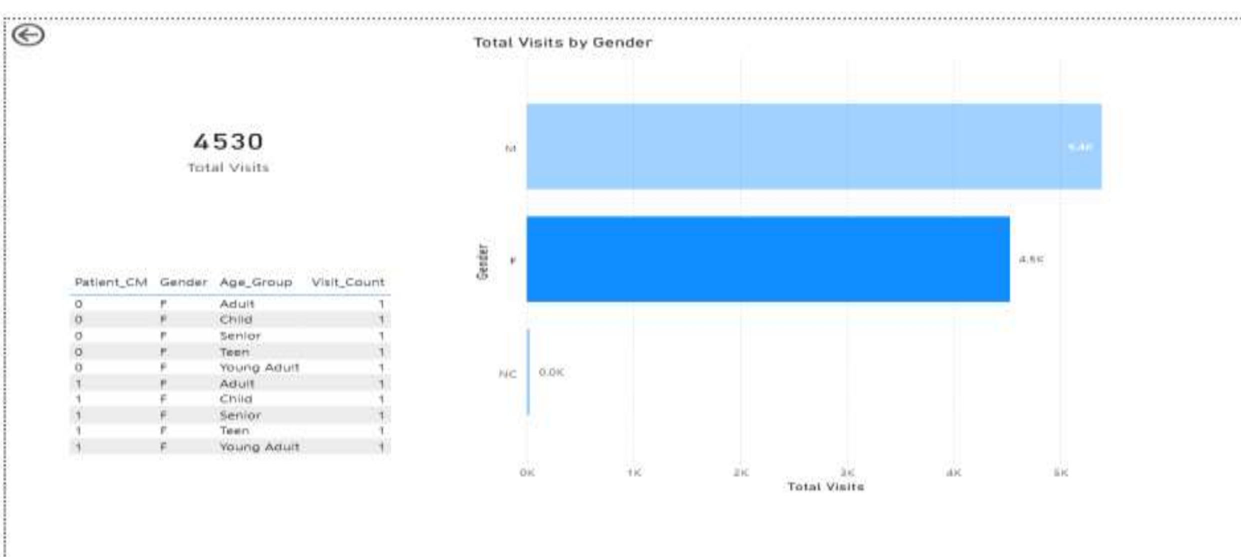


Figure 26: Drill-through for Another Department

DAX Measures used

The following DAX measures were used to support analysis:

- **Total Visits**
  = SUM(Fact_ER_Visit[Visit_Count])
- **Average Wait Time**
  = AVERAGE(Fact_ER_Visit[Wait_Time])
- **Average Satisfaction Score**
  = AVERAGE(Fact_ER_Visit[Satisfaction_Score])

Data Preparation and Modeling

Data preparation was performed by connecting Power BI to the SSAS cube created from the data warehouse. The data model follows a star schema structure, where the central fact table is connected to dimension tables.

This structure ensures efficient querying and enables multidimensional analysis using different attributes such as time, department, and patient information.

Visual Design

Various visualizations such as bar charts, line charts, pie charts, and card visuals were used to represent the data effectively. The layout was designed to improve readability and provide an interactive user experience.

Appropriate visual types were selected based on the nature of the data to ensure clarity and better understanding.

Power BI Service Publishing

The developed Power BI reports were published to the Power BI Service to enable online access and sharing.

The report was successfully uploaded, and all visuals including slicers, drill-down, and drill-through features were functioning correctly in the Power BI Service environment.

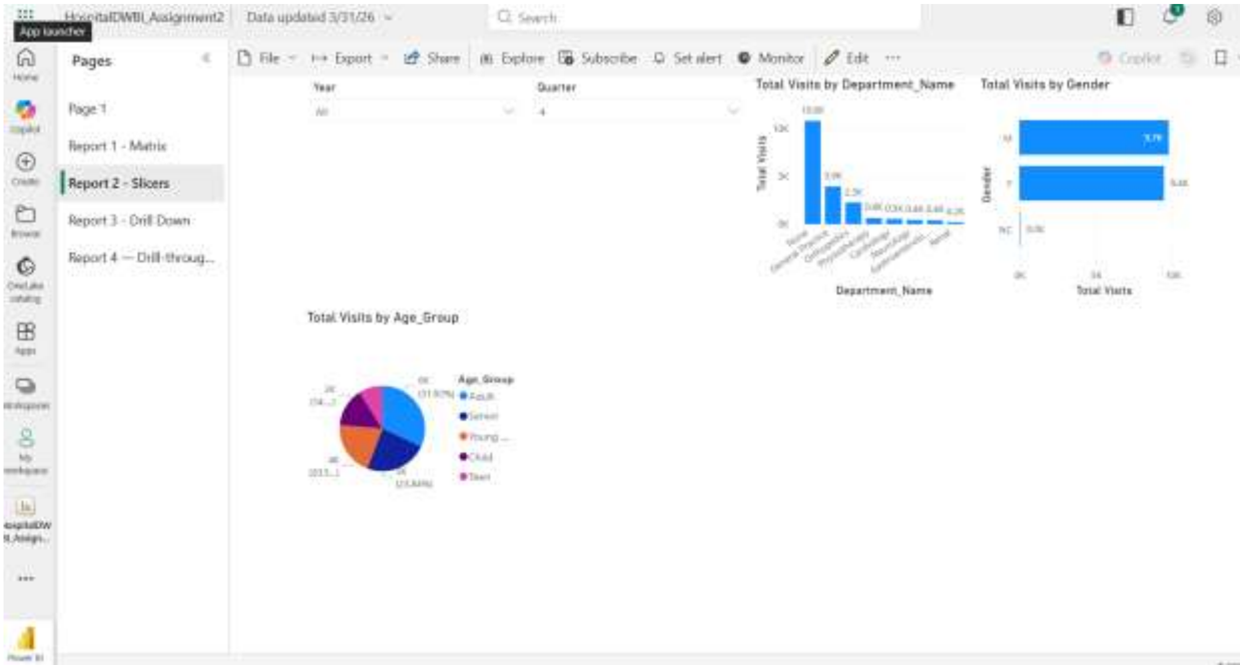


Figure 27: Published Report in Power BI Service

Conclusion

This assignment successfully demonstrated the implementation of a data warehouse, SSAS cube, OLAP operations, and Power BI reports.

The use of interactive dashboards and analytical techniques helped to identify trends and patterns in emergency room operations, supporting better decision-making and data-driven insights.

References

- Microsoft Corporation. (n.d.). *Create a Multidimensional Model by using SQL Server Data Tools (SSDT)*. Microsoft Learn. Retrieved from https://learn.microsoft.com/en-us/analysisservices/multidimensional-models/
- Microsoft Corporation. (n.d.). *Power BI documentation*. Microsoft Learn. Retrieved from https://learn.microsoft.com/en-us/power-bi/

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